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Analytical Report of the Second Assignment

AERONAUTICAL ENGINEERING - STRUCTURAL DYNAMICS AND AEROELASTICITY COURSE

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1. Introduction

The system under investigation is a straight, uniform semi-wing of span b and chord c , perfectly clamped at the root ($y = 0$) and free at the tip ($y = b$). The Elastic Axis (EA) is located at mid-chord. The Aerodynamic Center (AC) is located forward of the elastic axis by a constant distance $e = c/4$. Regarding inertial properties, the wing has a mass per unit span m and a mass moment of inertia per unit span J , calculated with respect to the cross-section's Center of Gravity (CG). The spanwise center-of-gravity axis is located a constant distance $d = 0.15c$ behind the elastic axis.

The structure includes:

- a strut with axial stiffness EA and negligible mass, connecting the wind tunnel wall (Point A) to the wing midspan ($y_B = b/2$, Point B). The strut forms an angle γ with the wing longitudinal axis, and its chordwise attachment point on the airfoil can be varied along the coordinate x_B ;
- an additional external tank modeled as a concentrated point mass M_T located at point (y_T, z_T) . The tank's rotational inertia is neglected, and its chordwise position lies on the elastic axis ($x_{CG,T} = 0$);
- an aileron of span b_a located at the outboard portion of the semi-wing.

The system operates in a steady two-dimensional aerodynamic flow with an asymptotic freestream velocity U_∞ in standard atmosphere at zero altitude. It is subjected to both control forces (aileron deflection $\beta(t)$) and uniform vertical gust $v_G(t)$.

2. Kinematics

The aeroelastic behaviour of the system is described through two Degrees of Freedom (DoFs):

1. $w(y)$: vertical displacement of the Elastic Axis due to bending (defined positive upwards);
2. $\theta(y)$: torsional rotation of the cross-section about the Elastic Axis.

Under the assumptions of small displacements and rigid cross-sections, the vertical displacement of a generic point $P(x, y)$ on the wing can be expressed as:

$$u(x, y) = w(y) - x\theta(y) \quad (1)$$

where the chordwise coordinate x originates at the elastic axis and is positive towards the trailing edge.

The kinematics of the characteristic points involved in the formulation are derived directly from this displacement field.

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The strut attachment point B is located at the wing midspan ($y_B = b/2$) with chordwise coordinate x_B . Its vertical displacement is therefore:

$$u_B = w(y_B) - x_B \theta(y_B) \quad (2)$$

The Aerodynamic Center is positioned ahead of the Elastic Axis by a distance e , corresponding to the coordinate $x_{AC} = -e$. Its displacement field becomes:

$$u_{AC}(y) = w(y) + e \theta(y) \quad (3)$$

Similarly, the wing Center of Gravity is located aft of the Elastic Axis by the distance d ($x_{CG} = d$), leading to:

$$u_{CG}(y) = w(y) - d \theta(y) \quad (4)$$

The external tank is positioned at the spanwise location y_T , with zero chordwise offset ($x_T = 0$) and vertical offset z_T measured positively upward from the Elastic Axis. Its displacement is therefore written as:

$$u_{T, \text{Vertical}} = w(y_T) + z_T \quad (5)$$

for the vertical displacement, while for the horizontal one, the expression is:

$$u_{T, \text{Horizontal}} = z_T \theta(y_T) \quad (6)$$

2.1. Principle of Virtual Works (PVW)

Once the kinematics have been defined, the model development can proceed by writing the Principle of Virtual Work (PVW), stating that $0 = \delta W$ or, more precisely $\delta W_{\text{int}} = \delta W_{\text{inertia}} + \delta W_{\text{ext}}$. δW_{int} and δW_{ext} are respectively the internal and external virtual works while $\delta W_{\text{inertia}}$ is the inertial virtual work.

The internal virtual work is generated by the elastic strain energy stored during the bending and torsional deformation of the continuous wing, coupled with the concentrated energy contribution from the strut's axial deformation. The strut acts as a concentrated equivalent vertical spring at the midspan section ($y_B = b/2$). The total internal virtual work is formulated as:

$$\delta W_{\text{int}} = \int_0^b EJ w''(y) \delta w''(y) dy + \int_0^b GJ \theta'(y) \delta \theta'(y) dy + \delta W_{\text{strut}} \quad (7)$$

The strut contribution is derived by projecting its axial stiffness EA through the angle γ to define an equivalent vertical stiffness K_B :

$$K_B = \frac{EA}{L_{\text{strut}}} \sin^2(\gamma) \quad (8)$$

where the strut length is $L_{\text{strut}} = \frac{b/2}{\cos(\gamma)}$. The virtual work of the strut is given by $K_B u_B(t) \delta u_B$. Substituting the previously defined kinematics for point B (eq 2):

$$\delta W_{\text{strut}} = K_B \left[w\left(\frac{b}{2}, t\right) - x_B \theta\left(\frac{b}{2}, t\right) \right] \left[\delta w\left(\frac{b}{2}\right) - x_B \delta \theta\left(\frac{b}{2}\right) \right] \quad (9)$$

The inertial virtual work includes the distributed mass contribution of the semi-wing and the concentrated mass contribution of the external tank. For the distributed wing mass, the acceleration of a generic point $P(x, y)$ on the cross-section is $\ddot{u}(x, y, t) = \ddot{w}(y) - x \ddot{\theta}(y)$. Integrating over the wing cross-sectional area A yields the distributed inertial loads. The moment of inertia transported to the Elastic Axis is $J_{EA} = J + md^2$. The distributed inertial virtual work is:

$$\delta W_{\text{iner, wing}} = - \int_0^b \left[\left(m \ddot{w}(y) - md \ddot{\theta}(y) \right) \delta w(y) + \left((J + md^2) \ddot{\theta}(y) - md \ddot{w}(y) \right) \delta \theta(y) \right] dy. \quad (10)$$

For the concentrated tank mass M_T located at y_T , the acceleration is defined by the kinematics derived earlier: $\ddot{u}_T(t) = \ddot{w}(y_T) + z_T \ddot{\theta}(y_T)$. The virtual work contributed by the tank's inertia is:

$$\delta W_{\text{iner,tank}} = -M_T \ddot{w}(y_T) \delta w(y_T) - M_T z_T^2 \ddot{\theta}(y_T) \delta \theta(y_T) \quad (11)$$

The work of external forces is defined by the action of the distributed aerodynamic lift $L(y)$ and the aerodynamic moment with respect to the Elastic Axis $M_{\text{EA}}(y)$:

$$\delta W_{\text{ext}} = \int_0^b L(y) \delta w(y) dy + \int_0^b M_{\text{EA}}(y) \delta \theta(y) dy$$

The aerodynamic forces are modeled using quasi-steady strip theory, which implies they depend on the instantaneous effective angle of attack α_{eff} at the Aerodynamic Center. The angle of attack is modified by the structural kinematic velocities (plunge and pitch rates) which generate aerodynamic damping, and by the vertical gust $v_G(t)$. Using the velocity at the Aerodynamic Center the effective angle of attack is:

$$\alpha_{\text{eff}}(y) = \theta(y) - \frac{\dot{w}(y) + e\dot{\theta}(y) - v_G(t)}{U_\infty} \quad (12)$$

The aerodynamic loads can be formulated as:

$$L(y) = q_\infty c C_{L/\alpha} \alpha_{\text{eff}}(y) + q_\infty c C_{L/\beta} \beta(y) \quad (13)$$

$$M_{\text{EA}}(y) = e L(y) + q_\infty c^2 C_{MAC/\beta} \beta(y) \quad (14)$$

where $q_\infty = \frac{1}{2} \rho U_\infty^2$ is the dynamic pressure.

3. Ritz-Galerkin Method

To transform the continuous differential problem into a finite-dimensional algebraic system of ordinary differential equations (ODEs), the Ritz-Galerkin method is applied. The continuous displacement fields $w(y)$ and $\theta(y)$ are approximated using a linear combination of trial functions (shape functions) multiplied by time-dependent generalized coordinates:

$$w(y) \approx \sum_{i=1}^{N_w} \phi_i(y) q_{w,i}(t) = \boldsymbol{\phi}^T(y) \mathbf{q}_w(t) \quad (15)$$

$$\theta(y) \approx \sum_{j=1}^{N_\theta} \psi_j(y) q_{\theta,j}(t) = \boldsymbol{\psi}^T(y) \mathbf{q}_\theta(t) \quad (16)$$

The trial functions $\phi_i(y)$ and $\psi_j(y)$ must be kinematically admissible, meaning they must strictly satisfy the geometric boundary conditions at the clamped root of the wing. Specifically, $w(0) = 0$, $w'(0) = 0$, and $\theta(0) = 0$. A suitable choice of simple polynomial bases satisfying these constraints is:

$$\phi_i(y) = \left(\frac{y}{b}\right)^{i+1} \quad \text{for } i = 1, \dots, N_w \quad (17)$$

$$\psi_j(y) = \left(\frac{y}{b}\right)^j \quad \text{for } j = 1, \dots, N_\theta \quad (18)$$

By substituting the approximated fields and their virtual variations ($\delta w = \boldsymbol{\phi}^T \delta \mathbf{q}_w$ and $\delta \theta = \boldsymbol{\psi}^T \delta \mathbf{q}_\theta$) into the Principle of Virtual Works, and imposing the arbitrariness of the virtual variations, the continuous problem is reduced to the canonical second-order linear matrix equation:

$$\mathbf{M}_s \ddot{\mathbf{q}}(t) + \mathbf{C}_a \dot{\mathbf{q}}(t) + (\mathbf{K}_s - \mathbf{K}_a) \mathbf{q}(t) = \mathbf{F}_{\text{ail}}(t) + \mathbf{F}_{\text{gust}}(t) \quad (19)$$

where $\mathbf{q}(t) = \{\mathbf{q}_w^T, \mathbf{q}_\theta^T\}^T$ is the global vector of the generalized coordinates. The system matrices are assembled in a block-partitioned format, separating the bending and torsional degrees of freedom.

3.1. System Matrices Definition

Structural Mass Matrix ($\mathbf{M}_s$)

The structural mass matrix captures the inertial properties of the wing and the external tank. Due to the CG offset (d) and the tank's vertical offset (z_T), the mass matrix features inertial coupling between bending and torsion:

$$\mathbf{M}_s = \begin{bmatrix} \mathbf{M}_s^{ww} & \mathbf{M}_s^{w\theta} \\ \mathbf{M}_s^{\theta w} & \mathbf{M}_s^{\theta\theta} \end{bmatrix} \quad (20)$$

with the sub-blocks defined as:

$$M_{s,ij}^{ww} = \int_0^b m \phi_i(y) \phi_j(y) dy + M_T \phi_i(y_T) \phi_j(y_T) \quad (21)$$

$$M_{s,ij}^{w\theta} = - \int_0^b m d \phi_i(y) \psi_j(y) dy \quad (22)$$

$$M_{s,ij}^{\theta w} = (M_{s,ji}^{w\theta})^T \quad (23)$$

$$M_{s,ij}^{\theta\theta} = \int_0^b (J + m d^2) \psi_i(y) \psi_j(y) dy + M_T z_T^2 \psi_i(y_T) \psi_j(y_T) \quad (24)$$

Structural Stiffness Matrix ($\mathbf{K}_s$)

The structural stiffness matrix accounts for the wing's internal elastic strain energy and the concentrated stiffness of the strut. The chordwise position of the strut (x_B) introduces elastic coupling:

$$\mathbf{K}_s = \begin{bmatrix} \mathbf{K}_s^{ww} & \mathbf{K}_s^{w\theta} \\ \mathbf{K}_s^{\theta w} & \mathbf{K}_s^{\theta\theta} \end{bmatrix} \quad (25)$$

$$K_{s,ij}^{ww} = \int_0^b E J \phi_i''(y) \phi_j''(y) dy + K_B \phi_i(b/2) \phi_j(b/2) \quad (26)$$

$$K_{s,ij}^{w\theta} = -K_B x_B \phi_i(b/2) \psi_j(b/2) \quad (27)$$

$$K_{s,ij}^{\theta w} = (K_{s,ji}^{w\theta})^T \quad (28)$$

$$K_{s,ij}^{\theta\theta} = \int_0^b G J \psi_i'(y) \psi_j'(y) dy + K_B x_B^2 \psi_i(b/2) \psi_j(b/2) \quad (29)$$

Aerodynamic Damping Matrix ($\mathbf{C}_a$)

The aerodynamic damping matrix arises from the structural velocity terms ($\dot{w}$ and $\dot{\theta}$) in the effective angle of attack formulation. When moved to the left-hand side of the dynamic equation, these terms yield a positive damping contribution:

$$\mathbf{C}_a = \frac{q_\infty c C_{L,\alpha}}{U_\infty} \begin{bmatrix} \mathbf{C}_a^{ww} & \mathbf{C}_a^{w\theta} \\ \mathbf{C}_a^{\theta w} & \mathbf{C}_a^{\theta\theta} \end{bmatrix} \quad (30)$$

$$C_{a,ij}^{ww} = \int_0^b \phi_i(y) \phi_j(y) dy \quad (31)$$

$$C_{a,ij}^{w\theta} = \int_0^b e \phi_i(y) \psi_j(y) dy \quad (32)$$

$$C_{a,ij}^{\theta w} = \int_0^b e \psi_i(y) \phi_j(y) dy \quad (33)$$

$$C_{a,ij}^{\theta\theta} = \int_0^b e^2 \psi_i(y) \psi_j(y) dy \quad (34)$$

Aerodynamic Stiffness Matrix ($\mathbf{K}_a$)

The aerodynamic stiffness matrix represents the self-excited loads strictly proportional to the geometric torsional rotation $\theta(y)$. Note that pure bending displacement w does not inherently generate aerodynamic forces in this strip theory formulation, hence the zero-block entries:

$$\mathbf{K}_a = q_\infty c C_{L,\alpha} \begin{bmatrix} \mathbf{0} & \mathbf{K}_a^{w\theta} \\ \mathbf{0} & \mathbf{K}_a^{\theta\theta} \end{bmatrix} \quad (35)$$

$$K_{a,ij}^{ww} = 0 \quad (36)$$

$$K_{a,ij}^{w\theta} = \int_0^b \phi_i(y) \psi_j(y) dy \quad (37)$$

$$K_{a,ij}^{\theta w} = 0 \quad (38)$$

$$K_{a,ij}^{\theta\theta} = \int_0^b e \psi_i(y) \psi_j(y) dy \quad (39)$$

Generalized Forcing Vectors ($\mathbf{F}_{\text{ail}}$, $\mathbf{F}_{\text{gust}}$)

The right-hand side of the governing equations contains the generalized forcing vectors associated with the external excitations. The aileron forces are integrated exclusively over the spanwise domain of the control surface $[b - b_a, b]$, driven by the aileron deflection time-history $\beta(t)$:

$$\mathbf{F}_{\text{ail}}(t) = \beta(t) \begin{Bmatrix} \mathbf{F}_{\text{ail}}^w \\ \mathbf{F}_{\text{ail}}^\theta \end{Bmatrix} \implies \begin{cases} F_{\text{ail},i}^w = q_\infty c C_{L,\beta} \int_{b-b_a}^b \phi_i(y) dy \\ F_{\text{ail},j}^\theta = q_\infty c (e C_{L,\beta} + c C_{\text{MAC},\beta}) \int_{b-b_a}^b \psi_j(y) dy \end{cases} \quad (40)$$

Conversely, the vertical gust $v_G(t)$ acts uniformly along the entire wing span $[0, b]$:

$$\mathbf{F}_{\text{gust}}(t) = \frac{v_G(t)}{U_\infty} \begin{Bmatrix} \mathbf{F}_{\text{gust}}^w \\ \mathbf{F}_{\text{gust}}^\theta \end{Bmatrix} \implies \begin{cases} F_{\text{gust},i}^w = q_\infty c C_{L,\alpha} \int_0^b \phi_i(y) dy \\ F_{\text{gust},j}^\theta = q_\infty e c C_{L,\alpha} \int_0^b \psi_j(y) dy \end{cases} \quad (41)$$

The formulation of this complete algebraic system serves as the foundational mathematical framework required to carry out all subsequent dynamic, frequency response, flutter, and stochastic analyses.

4. Convergence Study

To study the structural frequencies and complete the convergence analysis prior to the inclusion of aerodynamic forces, the free undamped vibration problem is considered ($U_\infty = 0$, leading to $\mathbf{C}_a = \mathbf{0}$ and $\mathbf{K}_a = \mathbf{0}$). The governing matrix equation reduces to:

$$\mathbf{M}_s \ddot{\mathbf{q}}(t) + \mathbf{K}_s \mathbf{q}(t) = \mathbf{0} \quad (42)$$

By assuming a harmonic solution of the form $\mathbf{q}(t) = \hat{\mathbf{q}} e^{j\omega t}$, where ω denotes the natural circular frequency and $\hat{\mathbf{q}}$ represents the mode shape vector of generalized coordinates, we obtain the standard linear eigenvalue problem:

$$(\mathbf{K}_s - \omega^2 \mathbf{M}_s) \hat{\mathbf{q}} = \mathbf{0} \quad (43)$$

The non-trivial solutions are found by enforcing the characteristic equation $\det(\mathbf{K}_s - \omega^2 \mathbf{M}_s) = 0$. This yields a set of discrete eigenvalues ω_n^2 and corresponding eigenvectors $\hat{\mathbf{q}}_n$.

To determine the minimum number of trial functions N_w and N_θ required for spatial convergence, an iterative process is established. The convergence rate is monitored via the relative error metric ϵ evaluated on the first bending frequency (ω_{b1}) and the first torsional frequency ($\omega_{\theta 1}$):

$$\epsilon = \frac{|\omega_n^{(N)} - \omega_n^{(N-1)}|}{\omega_n^{(N)}} < 10^{-3} \quad (44)$$

where $N = N_w = N_\theta$ represents the order of the polynomial expansion. The system is systematically solved by increasing N until condition (44) is fully satisfied for both target modes simultaneously.

5. Frequency Response Function (FRF) at Point P

For a steady forward flight speed of $U_\infty = 25$ m/s, the wing is subjected to a purely harmonic control input via the aileron deflection $\beta(t) = \beta_0 e^{j\Omega t}$, where Ω is the excitation frequency. The governing aeroelastic system of equations becomes:

$$\mathbf{M}_s \ddot{\mathbf{q}}(t) + \mathbf{C}_a \dot{\mathbf{q}}(t) + (\mathbf{K}_s - \mathbf{K}_a) \mathbf{q}(t) = \mathbf{F}_{\text{ail}}^0 \beta_0 e^{j\Omega t} \quad (45)$$

where $\mathbf{F}_{\text{ail}}^0 = \{\mathbf{F}_{\text{ail}}^w, \mathbf{F}_{\text{ail}}^\theta\}^T$ is the amplitude vector defined in the previous section. Seeking a harmonic state response $\mathbf{q}(t) = \hat{\mathbf{q}} e^{j\Omega t}$, the algebraic transfer function relating the generalized coordinates to the input amplitude is explicitly derived as:

$$\mathbf{H}_q(\Omega) = \frac{\hat{\mathbf{q}}}{\beta_0} = [-\Omega^2 \mathbf{M}_s + j\Omega \mathbf{C}_a + (\mathbf{K}_s - \mathbf{K}_a)]^{-1} \mathbf{F}_{\text{ail}}^0 \quad (46)$$

The targeted physical output is the vertical displacement of point P, located at the wing tip ($y_P = b$) and on the leading edge. According to the reference system configuration, the chordwise coordinate of the leading edge is $x_P = -c/2$. Substituting this position into the kinematic displacement field yields:

$$u_P = w(b) - \left(-\frac{c}{2}\right) \theta(b) = w(b) + \frac{c}{2} \theta(b) \quad (47)$$

Projecting this relation onto the Ritz trial space, the physical displacement can be expressed through an output matrix calculation:

$$u_P = \boldsymbol{\phi}^T(b) \mathbf{q}_w(t) + \frac{c}{2} \boldsymbol{\psi}^T(b) \mathbf{q}_\theta(t) = \mathbf{C}_P \mathbf{q}(t) \quad (48)$$

where the output row vector is structured as $\mathbf{C}_P = [\boldsymbol{\phi}^T(b) \quad \frac{c}{2} \boldsymbol{\psi}^T(b)]$. Consequently, the Frequency Response Function $H_P(\Omega)$ linking the aileron excitation to the structural displacement at point P is defined as:

$$H_P(\Omega) = \mathbf{C}_P \mathbf{H}_q(\Omega) = \mathbf{C}_P [-\Omega^2 \mathbf{M}_s + j\Omega \mathbf{C}_a + (\mathbf{K}_s - \mathbf{K}_a)]^{-1} \mathbf{F}_{\text{ail}}^0 \quad (49)$$

The magnitude and phase of $H_P(\Omega)$ are plotted versus Ω to construct the Bode diagram of the aeroelastic system.

6. Dynamic Stability

To investigate the onset of aeroelastic flutter, the homogeneous, self-excited system is rewritten in first-order state-space variables. Defining the state vector as $\mathbf{x}(t) = \{\mathbf{q}(t)^T, \dot{\mathbf{q}}(t)^T\}^T$, the state equation yields:

$$\dot{\mathbf{x}}(t) = \mathbf{A}(U_\infty) \mathbf{x}(t) \quad (50)$$

where the parameter-dependent state matrix $\mathbf{A}(U_\infty)$ is partitioned as:

$$\mathbf{A}(U_\infty) = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}_s^{-1} (\mathbf{K}_s - \mathbf{K}_a(U_\infty)) & -\mathbf{M}_s^{-1} \mathbf{C}_a(U_\infty) \end{bmatrix} \quad (51)$$

Recalling that $\mathbf{C}_a \propto U_\infty$ and $\mathbf{K}_a \propto U_\infty^2$, the stability of the system is tracked by computing the complex eigenvalues $\lambda_k = \sigma_k \pm j\omega_k$ of $\mathbf{A}$ across an increasing sweep of the freestream velocity U_∞ . The flutter speed V_F is defined as the lowest velocity at which the real part of any eigenvalue pairs transitions from negative to positive while maintaining a non-zero imaginary part:

$$V_F = \inf \left\{ U_\infty > 0 \mid \max_k (\text{Re}(\lambda_k(U_\infty))) = 0 \quad \text{and} \quad \text{Im}(\lambda_k(U_\infty)) \neq 0 \right\} \quad (52)$$

Initially, V_F is evaluated at the nominal strut configuration $x_B = -0.5c$. Subsequently, to find the chordwise position of the attachment point that minimizes the flutter speed, an optimization problem is formulated:

$$\min_{x_B} V_F(x_B) \quad \text{subject to } x_B \in [-0.5c, 0.5c] \quad (53)$$

This parametric study highlights the high sensitivity of the aeroelastic coupling mechanisms to the chordwise placement of the elastic constraints.

7. Gust Response with Direct Recovery

The system is next subjected to a sharp-edged deterministic vertical gust represented as a step input $v_G(t) = v_0 \mathcal{H}(t)$, where $\mathcal{H}(t)$ is the Heaviside step function. The operating flight speed is fixed at a subcritical condition $U_\infty = 0.75V_F$. The state-space system equations are supplemented by the gust forcing vector:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}_{\text{gust}}v_G(t), \quad \mathbf{B}_{\text{gust}} = \left\{ \frac{1}{U_\infty} \mathbf{M}_s^{-1} \mathbf{F}_{\text{gust}}^0 \right\} \quad (54)$$

The time response of the generalized coordinates $\mathbf{q}(t)$ is obtained by integrating the state-space equations numerically or analytically using the matrix exponential. To evaluate the structural demand, the internal root bending moment $M_x(0, t)$ at the clamp ($y = 0$) is computed via the direct recovery approach (Mode Displacement method). Based on Euler-Bernoulli beam theory, the internal bending moment relates directly to the curvature of the bending deflection:

$$M_x(0, t) = EJw''(0, t) = EJ \sum_{i=1}^{N_w} \phi_i''(0) q_{w,i}(t) \quad (55)$$

Recalling the chosen polynomial basis functions $\phi_i(y) = (y/b)^{i+1}$, their second spatial derivatives evaluate to:

$$\phi_i''(y) = \frac{(i+1)i}{b^2} \left(\frac{y}{b} \right)^{i-1} \quad (56)$$

Evaluating this expression at the root ($y = 0$), it is observed that for $i = 1$, $\phi_1''(0) = 2/b^2$, whereas for all higher orders $i > 1$, $\phi_i''(0) = 0$. Consequently, the direct structural recovery formula simplifies to a direct scaling of the first bending generalized coordinate:

$$M_x(0, t) = \frac{2EJ}{b^2} q_{w,1}(t) \quad (57)$$

8. White Noise Gust Input and Lyapunov Equation

Finally, the atmospheric turbulence is modeled as a stochastic process where the uniform vertical gust $v_G(t)$ is assumed to be a stationary Gaussian white noise with unit intensity, such that its autocorrelation function satisfies $E[v_G(t)v_G(t+\tau)] = \delta(\tau)$.

Operating at the same subcritical speed ($U_\infty = 0.75V_F$), the state covariance matrix $\sigma_{xx}^2 = E[\mathbf{x}(t)\mathbf{x}(t)^T]$ captures the statistical properties of the structural response. Since the system is linear and asymptotically stable at this velocity, the steady-state covariance matrix is found directly by solving the continuous Lyapunov algebraic equation:

$$\mathbf{A}\sigma_{xx}^2 + \sigma_{xx}^2\mathbf{A}^T + \mathbf{B}\mathbf{W}\mathbf{B}^T = \mathbf{0} \quad (58)$$

where $\mathbf{W} = [1]$ represents the spectral density of the white noise input.

Once σ_{xx}^2 is computed, the statistical variance of the root bending moment $\sigma_{M_x}^2$ is obtained. Expressing the root bending moment as a linear combination of the state variables via an output vector $\mathbf{C}_M$:

$$M_x(0, t) = \mathbf{C}_M\mathbf{x}(t), \quad \mathbf{C}_M = \left[\frac{2EJ}{b^2} \quad 0 \quad \dots \quad 0 \right] \quad (59)$$

the variance $\sigma_{M_x}^2$ is computed by applying the covariance propagation law:

$$\sigma_{M_x}^2 = \mathbf{C}_M\sigma_{xx}^2\mathbf{C}_M^T \quad (60)$$

This completes the full analytical formulation required to solve the assignment objectives.